…………………….Lecture One ………………………………..

1 . Solubility in quantitative terms as the concentration of solute in
A . a sub saturated solution at a certain temperature.
B . a saturated solution at a certain temperature.
C . a saturated solution at low temperature.
D . a saturated solution at high temperature.

2 . Solubility in a qualitative way, the spontaneous interaction of two or more substances to form
A . a homogeneous molecular dispersion.
B . a heterogeneous molecular dispersion.
C . complexation.
D . none of above.

3 . Solubility is an intrinsic material property that can be altered only by
A . vaporization.
B . decomposition.
C . physical modification of the molecule.
D . chemical modification of the molecule.

4 . solubility of a compound depends on
A . chemical properties only of the solute and the solvent.
B . physical properties only of the solute and the solvent.
C . physical and chemical properties of the solute and the solvent.
D . physical and chemical properties of the solute only.

5 . Solubility, the phenomenon of dissolution of solute in solvent to give
A . a homogenous system.
B . a heterogeneous system.
C . a mixture.
D . an equilibrium system.

6 . More than 40% NCEs (new chemical entities) developed in pharmaceutical industry are
A . practically insoluble in water.
B . Very soluble in water.
C . Freely soluble in water.
D . Soluble in water.

7 . Any drug to be absorbed must be present in
A . the form of colloidal at the site of absorption.
B . the form of solute at the site of absorption.

C . the form of solvent at the site of absorption.
D . the form of solution at the site of absorption.

8 . Various techniques are used for the enhancement of the solubility of poorly soluble drugs like
A . particle size reduction , crystal engineering.
B . salt formation, solid dispersion.
C . use of surfactant, complexation.
D . all of above.

9 . Selection of solubility improving method depends on
A . drug property.
B . site of absorption.
C . required dosage form characteristics.
D . all of above.

10 . A saturated solution is one in which
A . the solute in solution is in equilibrium with the solid phase.
B . the solvent in solution is in equilibrium with the solid phase.
C . the solute in solution is in equilibrium with the liquid phase.
D . the solute in solution is in equilibrium with the gas phase.

11 . An unsaturated or sub saturated solution is one containing the dissolved solute in a concentration below that necessary for
A . partial saturation at a definite temperature.
B . complete saturation at a definite temperature.
C . complete saturation at low temperature.
D . complete saturation at high temperature.

12 . A supersaturated solution is one that contains more of the dissolved solute than it would normally contain at
A . high temperature, were the un dissolved solute present.
B . low temperature, were the un dissolved solute present.
C . a definite temperature, were the un dissolved solute present.
D . moderate temperature, were the un dissolved solute present.

13 . The intermolecular forces that determine thermodynamic solubility
A . Solvent and solute are segregated.
B . To move a solute molecule into solution.
C . Once the solute molecule is surrounded by solvent.
D . all of above.

14 . structural effects in solubility that lead to mutual interactions between
A . solute and the solvent.
B . solvent and solvent.
C . solute and solute.
D . none of above.

15 . Solubility is quantitatively expressed in terms of
A . molarity.
B . normality.
C . percentage.
D . all of above.

16 . dipole moments alone
A . is not adequate to explain the solubility of nonpolar substances in water.
B . is adequate to explain the solubility of polar substances in water.
C . is not adequate to explain the solubility of polar substances in water.
D . is not adequate to explain the solubility of semi polar substances in water.

17 . As the length of a nonpolar chain of an aliphatic alcohol increases
A . the solubility of the compound in water don't change.
B . the solubility of the compound in water decreases.
C . the solubility of the compound in water increases.
D . none of above.

18 . Straight-chain monohydroxy alcohols with more than four or five carbons
A . cannot enter into the hydrogen-bonded structure of water and hence are only very soluble.
B . can enter into the hydrogen-bonded structure of water and hence are only slightly soluble.
C . cannot enter into the hydrogen-bonded structure of water and hence are only slightly soluble.
D . cannot enter into the hydrogen-bonded structure of water and hence are only freely soluble.

19 . additional polar groups are present in the molecule
A . water solubility less decreases .
B . water solubility less increases .
C . water solubility decreases greatly.
D . water solubility increases greatly.

20 . Branching of the carbon chain
A . reduces the nonpolar effect and leads to increased water solubility.
B . increase the nonpolar effect and leads to increased water solubility.
C . reduces the nonpolar effect and leads to decreased water solubility.

D . don’t change water solubility.

21 . Tertiary butyl alcohol is miscible in all proportions with water, whereas n-butyl alcohol dissolves to the extent of about
A .18 g/100 mL of water at 20°C.
B . 8 g/100 mL of water at 20°C.
C . 15 g/100 mL of water at 20°C.
D . 1 g/100 mL of water at 20°C.
22 . Nonpolar solvents are
A . able to reduce the attraction between the ions of strong and weak electrolytes.
B . unable to reduce the attraction between the ions of strong and weak electrolytes.
C . unable to increase the attraction between the ions of strong and weak electrolytes.
D . unable to reduce the attraction between the ions of non electrolytes.

23 . Nonpolar solvents are unable to reduce the attraction between the ions of strong and weak electrolytes
A . because of the solvent's high dielectric constants .
B . because of the solvent's high temperature.
C . because of the solvent's low dielectric constants .
D . because of the solvent's low temperature.

24 . Nor can the solvents break covalent bonds and ionize weak electrolytes, because
A . they belong to the group known as aprotic solvents.
B . they belong to the group known as protic solvents.
C . they belong to the group known as semi polar solvents.
D . none of the above.

25 . ionic and polar solutes are
A . not soluble or are only slightly soluble in polar solvents.
B . not soluble or are only freely soluble in nonpolar solvents.
C . soluble or are only slightly soluble in nonpolar solvents.
D . not soluble or are only slightly soluble in nonpolar solvents.
26 . Nonpolar compounds, however, can dissolve nonpolar solutes with
A . similar internal pressures through induced dipole interactions.
B . different internal pressures through induced dipole interactions.
C . similar internal pressures through cohesive forces.
D . similar internal pressures through adhesive forces.

27 . solute molecules are kept in solution by
A . the cohesive forces.
B . the weak van der Waals– London type of forces.

C . the weak hydrogen bonding.
D . the adhesive forces.

28 . oils and fats dissolve in
A . carbon tetrachloride.
B . benzene.
C . mineral oil.
D . all of above .

29 . Alkaloidal bases and fatty acids also dissolve in
A . nonpolar solvents.
B . polar solvents
C . water
D . semi polar solvent .

30 . semipolar compounds can act as
A . nonpolar solvents to bring about miscibility of polar and nonpolar liquids.
B . polar solvents to bring about miscibility of polar and nonpolar liquids.
C . intermediate solvents to bring about miscibility of polar and nonpolar liquids.
D . none of above

31 . acetone
A . increases partially the solubility of ether in water.
B . decreases the solubility of ether in water.
C . increases partially the solubility of ether in water.
D . increases the solubility of ether in water.

31 . Loran and Guth studied the intermediate solvent action of alcohol on
A . water–oleic acid mixtures.
B . water–mineral oil mixtures.
C . water–castor oil mixtures.
D . water– vegetables oils mixtures.

32 . Propylene glycol has been shown to
A . increase the mutual solubility of acetone and peppermint oil and of water and benzyl benzoate.
B . decrease the mutual solubility of water and peppermint oil and of water and benzyl benzoate.
C . increase the mutual solubility of water and peppermint oil and of water and benzyl benzoate.
D . increase the mutual solubility of water and peppermint oil and of acetone and benzyl benzoate.

33.Propylparaben is
A . Very poorly soluble in water
B . Highly soluble in water
C . soluble in both water as well as organic solvents
D . partially soluble in organic solvent

34. Freely soluble (FS)
A . From 1 to 10 Parts of Solvent Required for One Part of Solute.
B . From 10 to 30 Parts of Solvent Required for One Part of Solute.
C . From 30 to 100 Parts of Solvent Required for One Part of Solute.
D . From 100 to 1000 Parts of Solvent Required for One Part of Solute.

35. Sparingly soluble (SPS)
A . From 1 to 10 Parts of Solvent Required for One Part of Solute.
B . From 10 to 30 Parts of Solvent Required for One Part of Solute.
C . From 30 to 100 Parts of Solvent Required for One Part of Solute.
D . From 100 to 1000 Parts of Solvent Required for One Part of Solute.

36. Slightly soluble (SS)
A . From 1 to 10 Parts of Solvent Required for One Part of Solute.
B . From 10 to 30 Parts of Solvent Required for One Part of Solute.
C . From 30 to 100 Parts of Solvent Required for One Part of Solute.
D . From 100 to 1000 Parts of Solvent Required for One Part of Solute.

37. Very slightly soluble (VSS)
A . From1000 to10000 Parts of Solvent Required for One Part of Solute.
B . From 10 to 30 Parts of Solvent Required for One Part of Solute.
C . From 30 to 100 Parts of Solvent Required for One Part of Solute.
D . From 100 to 1000 Parts of Solvent Required for One Part of Solute.

38. There are ………… types of solvents
A . Four.
B . five.
C . three.
D . six.

39 .The component present in solution in small quantity is known as…..
A . Solvent
B . Solution
C . Solute
D . Liquid

40 . The dielectric constant of water is
A . 20
B . 5
C . 50
D . 80

……………………… Lecture Two ………………………………

41 . volatile oils are mixed with water to form
A . dilute solutions known as aromatic alcohols
B . dilute solutions known as aliphatic waters
C .concentrated solutions known as aromatic waters
D . dilute solutions known as aromatic waters

42 . Liquid–liquid systems can be divided into two categories according to the solubility of the substances in one another:
A . complete miscibility and partial miscibility.
B . complete miscibility and immiscibility.
C . concentrated and diluted.
D . saturated and unsaturated.

43 . The term miscibility refers to the
A . mutual solubilities of the components in liquid–liquid systems.
B . mutual solubilities of the components in gas–liquid systems.
C . mutual solubilities of the components in solid–liquid systems.
D . mutual solubilities of the components in solid–gas systems.

44 . effervescent preparations containing
A . oxygen that are dissolved and maintained in solution under positive pressure.
B . carbon dioxide that are dissolved and maintained in solution under negative pressure.
C . air that are dissolved and maintained in solution under positive pressure.
D . carbon dioxide that are dissolved and maintained in solution under positive pressure.

45 . Aerosol products in which the propellant is
A . either carbon monoxide or nitrogen.
B . either carbon dioxide or nitrogen.
C . either air or nitrogen.
D . only air.

46 . The solubility of a gas in a liquid is the concentration of the dissolved gas when it is
A . in equilibrium with some of the pure gas among the solution.
B . in equilibrium with some of the pure gas in the solution.
C . in equilibrium with some of the pure gas above the solution.

D . not in equilibrium with some of the pure gas above the solution.

47 . The solubility depends primarily on
A . the pressure.
B . presence of salts.
C . chemical reactions.
D . all of above.

48 . The effect of the pressure on the solubility of a gas is expressed by
A . Henry's law.
B . Rault law.
C . Boyle's law.
D . Charles's law

49 . Henry's law, states that in a very dilute solution at constant temperature,
A . the fraction mole of dissolved gas is proportional to the partial pressure of the gas above the solution at equilibrium .
B . the concentration of dissolved gas is proportional to the partial pressure of the gas above the solution at equilibrium .
C . the concentration of dissolved gas is proportional to the total pressure of the gas above the solution at equilibrium .
D . the concentration of dissolved gas is proportional to the temperature of the gas above the solution at equilibrium .

50 . In Henry's relationship $C2= \sigma p$, in which σ is sometimes referred to as the
A . temperature.
B . mole fraction.
C . solubility Coefficient.
D . length of the bond

51 . The significance or Henry's law for the pharmacist rests upon the fact that the solubility of a gas
A . increases directly with the volume of the gas.
B . increases directly with the temperature of the gas.
C . decreases directly with the pressure on the gas.
D . increases directly with the pressure on the gas.

52 . sometimes the gas escapes with violence when
A . the pressure above the solution is released.
B . the temperature is decreased.
C . the volume of the solution is increased.

D . the pressure of the solution is constant.

53 . Temperature also has a marked influence on the solubili1y of a
A . liquid in a liquid.
B . gas in a liquid.
C . solid in a liquid.
D . all of above

54. As the temperature increases, the solubili1y of most gases
A . remain constant
B . increases
C . decreases.
D . none of above .

55 . Gases are often liberated from solutions in which they are dissolved by
A . the introduction of oxygen.
B . the introduction of an electrolyte.
C . the introduction of acid.
D . the introduction of base.

56 . The salting out effect can be demonstrated by adding a
A . small amount of salt to a ‘carbonated’ solution.
B . large amount of salt to a ‘carbonated’ solution.
C . small amount of salt to saturated solution.
D . none of above .

57 . Henry's law applies strictly to gases that are only
A . very soluble in solution.
B . soluble in solution.
C . freely soluble in solution.
D . slightly soluble in solution.

58 . Gases such as hydrogen chloride, ammonia, and carbon dioxide show
A . precipitate as a result of chemical reaction between the gas and solvent.
B . deviations as a result of chemical reaction between the gas and solute.
C . deviations as a result of chemical reaction between the gas and solvent.
D . deviations as a result of physical change between the gas and solvent.

59 . deviations as a result of chemical reaction between the gas and solvent,
A . usually with a resultant increase in solubility.
B . usually with a resultant decrease in solubility.

C . usually with a resultant increase in pressure.
D . usually with a resultant increase in temperature.

60 . hydrogen chloride is about
A . I0 times more soluble in water than is oxygen.
B . 100 times more soluble in water than is oxygen.
C . I000 times more soluble in water than is oxygen.
D . I0.000 times more soluble in water than is oxygen.

61 . volatile oils are mixed with water to form dilute solutions known as
A . aliphatic water.
B . aromatic waters.
C . amines.
D . ketones or aldehydes.

62 . If one of the components shows a negative deviation, it can be demonstrated by the use of thermodynamics that the other component must
A . also show negative deviation.
B . don't change
C . show positive deviation.
D . none of above

63 . Negative deviation lead to
A . decreased solubility and are frequently associated with hydrogen bonding between nonpolar compounds.
B . increased solubility and are frequently associated with hydrogen bonding between nonpolar compounds
C . increased solubility and are frequently associated with hydrogen bonding between polar compounds
D . decreased solubility and are frequently associated with hydrogen bonding between polar compounds

64 . The interaction of the solvent with the solute is known as
A . absorption.
B . adsorption.
C . desorption.
D . solvation.

65 . Positive deviations leading to decreased solubility , are interpreted as resulting to
A . form double molecules (dimer) or polymer of higher order.
B . form complex.

C . form unstable compounds.
D . none of above.

66 . positive deviation is better accounted for in most cases by the difference in
A . the adhesive forces of the molecules of each constituent.
B . the cohesive forces of the molecules of each constituent.
C . dipole - dipole forces of the molecules of each constituent.
D . none of above.

67 . The attractive forces, which may occur in gases, liquids, or solids, are called
A . the adhesive forces of the molecules of each constituent.
B . the cohesive forces of the molecules of each constituent.
C . dipole - dipole forces of the molecules of each constituent.
D . internal pressure.

68 . When the vapor is assumed to be nearly ideal. the internal pressure in cal/cm^3 is obtained by using the equation
A . Pi = ΔSv-RT / V
B . Pi = ΔGv-RT / V
C . Pi = ΔHv-RT / V
D . Pi = ΔH-RT / V

69 . When the internal pressures or cohesive forces of the constituents of mixture such as hexane and water are quite different
A . the molecules of one constituents cannot mingle with those of the other, and partial solubility results
B . the molecules of one constituents cannot mingle with those of the other, and solubility results
C . the molecules of one constituents cannot mingle with those of the other, and complete solubility results
D . the molecules of one constituents cannot mingle with those of the other, and no solubility results

70 . Polar liquid have high cohesive forces that is
A . small internal pressures, and they are solvents only for compounds of similar nature.
B . large internal pressures, and they are solvents only for compounds of different nature.
C . large pressures, and they are solvents only for compounds of similar nature.
D . large internal pressures, and they are solvents only for compounds of similar nature.

71 . Nonpolar substances with low internal pressures are
A . squeezed out by the low attractive forces existing between the molecules of the polar liquid.

B . squeezed out by the powerful attractive forces existing between the molecules of the polar liquid.
C . get away by the powerful attractive forces existing between the molecules of the polar liquid.
D . squeezed out by the powerful attractive forces existing between the molecules of the nonpolar liquid.

72 . Many important drugs belong to the class of
A . weak acids and strong bases
B . strong acids and bases
C . weak acids and bases
D . strong acids and weak bases

73 . important drugs react with
A . strong acids and bases and, within definite ranges of pH, exist as ions that are ordinarily soluble in water.
B . weak acids and bases and, within definite ranges of pH, exist as ions that are ordinarily soluble in water.
C . strong acids and bases and, within definite ranges of pH, exist as ions that are ordinarily insoluble in water.
D . strong acids and bases and, within wide ranges of pH, exist as ions that are ordinarily soluble in water.

74 . The fatty acids containing
A . more than 7 carbon atoms form soluble soaps with the alkali metals.
B . more than 10 carbon atoms form soluble soaps with the alkali metals.
C . more than 10 carbon atoms form insoluble soaps with the alkali metals.
D . more than 10 carbon atoms form soluble soaps with the transition metals.

75 . carboxylic acids containing more than five carbons are react with
A . concentrated sodium hydroxide, carbonates, and bicarbonates to form soluble salts.
B . dilute sodium hydroxide, carbonates, and bicarbonates to form insoluble salts.
C . dilute sodium hydroxide, carbonates, and bicarbonates to form soluble salts.
D . dilute sodium hydroxide, carbonates, and bicarbonates to form Very slightly soluble salts.

76 . oleic acid ($C_{17}H_{33}COOH$) is insoluble in water but is soluble in
A . alcohol
B . ether
C . hydroxy acids
D . all of above

77 . tartaric and citric acids, are quite soluble in water because
A . they are solvated through their hydroxyl groups.
B . they are solvated through their carbonyl groups.
C . they are solvated through their amino groups.
D . they are solvated through their carboxylate groups.

78 . Sodium citrate is used sometimes to dissolve water-insoluble acetylsalicylic acid because
A . the effect of hydrogen bonding.
B . the soluble acetylsalicylic acid is formed in the reaction.
C . the soluble acetylsalicylate ion is formed in the reaction.
D . none of above

79 . the practice of dissolving aspirin is questionable because
A . the acetylsalicylate is also precipitate.
B . the acetylsalicylate is also hydrolyzed rapidly.
C . the acetylsalicylate is also hydrolyzed very slowly.
D . none of above.

80 . Aromatic acids react with
A . dilute alkalies to form water-soluble salts.
B . concentrated alkalies to form water-soluble salts.
C . dilute alkalies to form water-insoluble salts.
D . concentrated alkalies to form water-insoluble salts.

81 . Aromatic acids can be precipitated as the free acids if
A . oxidizing agent substances are added to the solution.
B . reducing agent substances are added to the solution.
C . weaker acidic substances are added to the solution.
D . stronger acidic substances are added to the solution.

82 . Aromatic acids can also be precipitated as
A . halogens should be added to the solution.
B . nonmetal ion should be added to the solution.
C . heavy metal salts, ions should be added to the solution.
D . none of above.

83 . Benzoic acid is soluble in
A . sodium hydroxide solution.
B . alcohol.
C . fixed oils.
D . all of above.

84 . The OH group of salicylic acid cannot contribute to the solubility because
A . it is involved in an intermolecular hydrogen bond.
B . it is involved in an intramolecular hydrogen bond.
C . it is involved in dipole-dipole interaction.
D . it is involved in Vander waals forces.

85 . Phenol is weakly acidic and only slightly soluble in water but is
A . quite soluble in dilute sodium hydroxide solution.
B . quite soluble in concentrated sodium hydroxide solution.
C . very soluble in dilute sodium hydroxide solution.
D . freely soluble in dilute sodium hydroxide solution.

86 . Phenol is a weaker acid than H_2CO_3 and is thus
A . displaced and precipitated by O_2 from its dilute alkali solution.
B . displaced and precipitated by N_2 from its dilute alkali solution.
C . displaced and precipitated by CO_2 from its dilute alkali solution.
D . displaced and precipitated by H_2 from its dilute alkali solution.

87 . carbonates and bicarbonates
A . cannot increase the solubility of phenols in water.
B . can increase the solubility of phenols in water.
C . partially increase the solubility of phenols in water.
D . decrease the solubility of phenols in water.

88 . Many organic compounds containing
A . hydrogen atom in the molecule are important in pharmacy.
B . oxygen atom in the molecule are important in pharmacy.
C . a basic nitrogen atom in the molecule are important in pharmacy.
D . carbon atom in the molecule are important in pharmacy.

89 . Most of organic compounds containing a basic nitrogen atom weak electrolytes are not very soluble in water but
A . are soluble in dilute solutions of acids.
B . are soluble in dilute solutions of bases.
C . are soluble in concentrated solutions of acids.
D . are soluble in concentrated solutions of bases.

90 . atropine sulfate and tetracaine hydrochloride are formed by
A . reacting the acidic compounds with bases.
B . reacting the basic compounds with acids.
C . reacting the reducing agents with acids.

D . reacting the oxidizing agents with acids.

91 . Addition of an alkali to a solution of the salt of atropine sulfate and tetracaine hydrochloride compounds
A . dissolve the free base from solution if the solubility of the base in water is low.
B . precipitates the free base from solution if the solubility of the base in water is high.
C . precipitates the free base from solution if the solubility of the base in water is low.
D . dissolve the free base from solution if the solubility of the base in water is high.

92 . The aliphatic nitrogen of the sulfonamides is sufficiently
A . negative so that these drugs act as slightly soluble weak acids rather than as bases.
B . positive so that these drugs act as slightly soluble weak acids rather than as bases.
C . negative so that these drugs act as very soluble weak acids rather than as bases.
D . negative so that these drugs act as insoluble weak acids rather than as bases.

93 . The oxygen's of the sulfonyl (—SO2—) group in sulfonamides
A . donate electrons, and the resulting electron deficiency of the sulfur atom.
B . withdraw electrons, and the resulting electron deficiency of the sulfur atom.
C . withdraw electrons, and the resulting electron rich of the sulfur atom.
D . none of above.

94 . The electron deficiency of the sulfur atom in sulfonamides results in the electrons of the N:H bond being
A . held more closely to the nitrogen atom. The hydrogen therefore is bound more firmly
B . held far away to the nitrogen atom. The hydrogen therefore is bound less firmly
C . held more closely to the nitrogen atom. The hydrogen therefore is bound less firmly
D . none of above.

95 . The sodium salts of the sulfonamides are precipitated from solution by the
A . addition of a weak acid or by a salt of a strong acid and a strong base such as ephedrine hydrochloride.
B . addition of a strong acid or by a salt of a strong acid and a strong base such as ephedrine hydrochloride.
C . addition of weak acid or by a salt of weak acid and a weak base such as ephedrine hydrochloride.
D . addition of a strong acid or by a salt of a strong acid and a weak base such as ephedrine hydrochloride.

96 . The barbiturates, like the sulfonamides, are weak acids because
A . the electronegative oxygen of each acidic carbonyl group tends to withdraw electrons and to create a positive carbon atom.

B . the electronegative oxygen of each acidic carbonyl group tends to donate electrons and to create a positive carbon atom.
C . the electronegative oxygen of each acidic carbonyl group tends to withdraw electrons and to create a negative carbon atom.
D . the electronegative oxygen of each acidic carbonyl group tends to withdraw electrons and to create a negative sulfur atom.

97 . in barbiturates the carbon in turn
A . attracts electrons from the nitrogen group and causes the hydrogen to be held more firmly.
B . attracts electrons from the nitrogen group and causes the hydrogen to be held less firmly.
C . repulsion electrons from the nitrogen group and causes the hydrogen to be held less firmly.
D . none of above.

98 . The barbiturates in sodium hydroxide solution, the hydrogen is readily lost, and the molecule
A . exists as a soluble cation of the weak acid.
B . exists as a soluble anion of the strong acid.
C . exists as a soluble anion of the weak acid.
D . exists as a soluble anion of the weak base.

……………….. Lecture Three……………………………………..

99 . The solubility of the salt refers to the mass of the salt which will dissolve per 100 mL of solvent (in this case, water) at
A . a particular enthalpy.
B . a particular temperature.
C . a particular pressure.
D . a particular volume.

100. Adding a common cation or anion shifts a solubility equilibrium in the direction predicted by
A . bronsted and lowry principle.
B . Lewis principle.
C . Henry's law.
D . Le Chatelier's principle

101. the solubility of any sparingly soluble salt is
A . almost always decreased by the presence of a soluble salt that contains high pH.
B . almost always decreased by the presence of a insoluble salt that contains a common ion.
C . almost always decreased by the presence of a soluble salt that contains a common ion.
D . almost always increased by the presence of a soluble salt that contains a common ion.

102. When slightly soluble electrolytes are dissolved to form saturated solutions,

A . the solubility is described by a special constant, known as the activity product.
B . the solubility is described by a special constant, known as the solubility product, K sp.
C . the solubility is described by a special constant, known as the activity coefficient.
D . the solubility is described by a special constant, known as the solubility equilibrium, K eq.

103. Silver chloride is an example of such a
A . freely soluble salt.
B . very soluble salt.
C . slightly soluble salt.
D . insoluble salt.

104. because the Silver chloride salt dissolves only with difficulty and
A . the ionic strength is high, the equilibrium expression can be written in terms of concentrations instead of activities.
B . the ionic strength is low, the equilibrium expression can be written in terms of concentrations instead of activities.
C . the ionic strength is low, the equilibrium expression can be written in terms of fraction mole instead of activities.
D . none of above.

105. The equation Ksp = [Ag+] [C1 -] is only approximate for
A . sparingly soluble salts.
B . very soluble salts.
C . insoluble salt.
D . freely soluble salt.

106. when activities rather than concentrations should be used in solubility product
A . It does hold for salts that are freely soluble water such as sodium chloride.
B . It does not hold for salts that are freely soluble water such as sodium chloride .
C . It does not hold for salts that are insoluble water such as sodium chloride.
D . It does not hold for salts that are Very slightly soluble water such as sodium chloride.

107. in the equation $Al(OH)_{3solid} \rightleftharpoons Al^{3+} + 3OH^-$
A . $[Al^{3+}]^3 [OH^-]^3$ = Ksp
B . $[Al^{3+}] [OH^-]^3$ = Ksp
C . $[Al^{3+}]^3 [OH^-]$ = Ksp
D . $[Al^{3+}] [OH^-]$ = Ksp

108. If an ion in common with AgCl, that is, Ag^+ or Cl^- is added to a solution of silver chloride,
A . there isn't an equilibrium.
B . the equilibrium is not change.

C . the equilibrium is altered.
D . none of above.

109. The addition of sodium chloride, to silver nitrate, increases the concentration of chloride ions so that momentarily
A . $[Ag^{+}][C1^{-}] > Ksp$
B . $[Ag^{+}][C1^{-}] < Ksp$
C . $[Ag^{+}][C1^{-}] = Ksp$
D . none of above.

110. the result of adding a common ion is
A . to kept the solubility of a slightly soluble electrolyte constant.
B . to increase the solubility of a slightly soluble electrolyte.
C . to reduce the solubility of a slightly soluble electrolyte.
D . none of above.

111. the common ion forms a complex with the salt whereby
A . kept the solubility constant.
B . the net solubility can be decreased.
C . the net solubility can be increased.
D . none of above.

112. Salts having no ion in common with the slightly soluble electrolyte produce an effect opposite to that of a common ion
A . At high concentration, they increase rather than decrease the solubility.
B . At moderate concentration, they increase rather than decrease the solubility.
C . At moderate concentration, they decrease rather than increase the solubility.
D . At low concentration, they increase rather than decrease the solubility.

113. the addition of an electrolyte that does not have an ion in common with AgCl
A . kept the solubility of silver chloride constant.
B . causes decrease in the solubility of silver chloride.
C . causes an increase in the solubility of silver chloride.
D . none of above.

114. A pharmacist wishing to prevent precipitation of a slightly soluble salt in water can add
A . some substance that will dissociate and reduce the concentration of one of the ions.
B . some substance that will tie up and increase the concentration of one of the ions.
C . some substance that will tie up and reduce the concentration of the two ions.
D . some substance that will tie up and reduce the concentration of one of the ions.

115. the addition of an electrolyte that does not have an ion in common with AgCl
A . kept the solubility of silver chloride constant.
B . causes decrease in the solubility of silver chloride.
C . causes an increase in the solubility of silver chloride.
D . none of above.

116. The sodium salts of the sulfonamides are precipitated from solution by
A . the addition of a strong acid.
B . or by a salt of a strong acid and a weak base such as ephedrine hydrochloride.
C . both A & B.
D . the addition of a strong base or by a salt of a strong base and a weak base such as ephedrine hydrochloride.

117. The barbiturates , in sodium hydroxide solution
A . The hydrogen is readily lost.
B . the molecule exists as a soluble anion of the weak acid.
C . both A & B.
D . The hydrogen is readily lost, and the molecule exists as a soluble cation of the weak acid.

118. in highly alkaline solution, of barbiturates,
A . the second hydrogen doesn't ionize.
B . the second hydrogen ionizes.
C . the second hydrogen bonded in hydrogen bonding.
D . none of above.

119. Although the barbiturates are soluble in alkalines,
A . they are precipitated as the free acids when a stronger acid is added.
B . and the pH of the solution is lowered.
C . both A & B.
D . they are precipitated as the free acids when a stronger acid is added and the pH of the solution is increased.

……………………..Lecture Four ……………….

120. 1 % solution of phenobarbital sodium is
A . soluble at pH values high in the alkaline range.
B . insoluble at pH values high in the alkaline range.
C . soluble at pH values low in the acidic range.
D . none of above .

121. The soluble ionic form is converted into molecular phenobarbital as
A . the pH is increased and reaches 10.3.
B . the pH is lowered and below 3.3.

C . the pH is lowered and below 8.3.
D . the pH is lowered and below 5.3.

122. The soluble ionic form is converted into molecular phenobarbital as
A . the pH is increased and reaches 10.3, the drug begins to converted from solution at room temperature .
B . the pH is lowered and below 8.3, the drug begins to converted from solution at low temperature .
C . the pH is lowered and below 8.3, the drug begins to converted from solution at room temperature .
D . the pH is lowered and below 8.3. the drug begins to converted from solution at high temperature .

123. alkaloidal salts such as atropine sulfate begin to
A . precipitate as the pH is constant .
B . precipitate as the pH is decreased.
C . ionize as the pH is elevated .
D . precipitate as the pH is elevated .

124. HP solution + H_2O $\longrightarrow$ H_3O^+ +p^-, Because the concentration of the un-ionized form in solution, HP solution is
A . essentially increased.
B . essentially constant.
C . essentially decreased.
D . essentially decreased at high temparture.

125. When the electrolyte is weak and does not dissociate appreciably, the solubility of the acid in water or acidic solutions is
A . So = (HP)
B . S = (HP)
C . So = (P^-).
D . S = (P^-).

126. What is USP?
A. The United States Pharmacology
B. The United States Pharmacy
C. The United States Pharmacopoeia
D. The United States Pharmaceuticals
127. So = (HP) which, for phenobarbital is approximately
A. 0.05 mole/liter.
B. 0.005 mole/liter.

C. 0. 5 mole/liter.
D. 5 mole/liter.

128. pHp = pKa + log S-So/So , where pHp depends on
A. the initial molar concentration, p- .of salt added.
B. the initial molar concentration, So .of salt added.
C. the initial molar concentration, S .of salt added.
D. the initial molar concentration, So & p- .of salt added..

129. pHp = pKa + log S-So/So , where pHp, is the pH below which the drug separates from solution as
A. the partially dissociated acid.
B. the dissociated acid.
C. the undissociated acid.
D. none of above.

130. the solubility of a weak base as a function of the pH of a solution. The expression is
A. pH p= pKw - pKb + log S–So/ So
B. pH p= pKw + log So/S – So
C. pH p= pKw + pKb + log So/S – So
D. pH p= pKw - pKb + log So/S – So

131. pH p= pKw - pKb + log So/S – So , where
A. S is the concentration of the drug initially added as the salt and So is the molar solubility of the free acid in water.
B. S is the concentration of the drug initially added as the salt and So is the molar solubility of the free base in water.
C. S is the concentration of the drug initially added as the solute and So is the molar solubility of the free base in water.
D. S is the concentration of the drug initially added as the solvent and So is the molar solubility of the free base in water.

132 . Elixirs are ……
A. Hydro alcoholic liquid
B. Aqueous
C. Viscous
D. Semi solids

133 . In $S_0 = HP_{sol}$, S_0 is
A. molar solubility
B. intrinsic solubility

C. both of these
D. none of above

134 .When the electrolyte is weak and does not dissociate appreciably, the solubility of the acid in water or acidic solutions is *S*o = [HP], which, for phenobarbital is approximately(in percentage)
A .0.10%
B .0.11%
C .0.12%
D .0.15%

135.$pH_p = pK_a + \log S - S_0 / S_0$ In this equation pH_p is
A . pH below which the drug separates from solution as the undissociated acid
B .pH above which the drug separates from solution as the undissociated acid
C .pH below which the drug separates from solution as the dissociated acid
D .all of above

136.$pH_p = pK_w - pK_b + \log S_0 / S - S_0$ In this equation S is
A . concentration of the drug initially added as the salt
B .molar solubility of the free base in water
C .intrinsic solubility
D .all of the above

137.$pH_p = pK_w - pK_b + \log S_0 / S - S_0$ In this equation So is
A .concentration of the drug initially added as the salt
B .molar solubility of the free base in water
C .intrinsic solubility
D .all of the above

138.$pH_p = pK_w - pK_b + \log S_0 / S - S_0$ In this equation pHp is
A .pH below which the drug separates from solution as the undissociated acid
B .pH above which the drug begins to precipitate from solution as the free base
C .pH below which the drug begins to precipitate from solution as the free base
D .all of these

139.Who studied the effect of an increase of alcohol concentration on the dissociation constant of sulfathiazole?
a)Stockton
b)Johnson
c)Higuchi
d)all of these

140.When the solution is of such a pH that the drug is entirely in the ionic form, it behaves as a solution of a strong electrolyte, and solubility does not constitute a serious problem.
A .true
B .false

141.When the pH is adjusted to a value at which ionized molecules are produced in sufficient concentration to exceed the solubility of this form, precipitation occurs.
A .true
B .false

142.Frequently, a solute is more soluble in a _____ than in one solvent alone.
A . mixture of solvents
B .basic solvent
C .acidic solvent
D .water alone

143.Cosolvency is
A .a solute is less soluble in a mixture of solvents than in one solvent alone
B .a solute is more soluble in a mixture of solvents than in one solvent alone
C .a solute is more soluble in one solvent alone than in mixture of solvents
D .all of these

144.The solvents that, in combination, increase the solubility of the solute are called
A .nonelectrolytes
B .electrolytes
C .cosolvents
D .dissociated ions

145.Approximately 1 g of phenobarbital is soluble in (at temperature 25°C)
A . 1000 mL of water
B . 10 mL of alcohol
C . 40 mL of chloroform
D . all of the above

146. Kramer and Aynn examined the solubility of hydrochloride salts of organic bases as a function of
A . pH
B . temperature
C . solvent composition
D . all of the above

147. Agarwal and Blake" and Schwartz et al. determined the solubility of phenytoin as a
A. temparture
B. pressure
C. mole fraction
D. none of above

148. Agarwal and Blake" and Schwartz et al. determined the solubility of phenytoin as a
A. function of pH and alcohol concentration in various buffer systems
B. calculated the apparent dissociation constant
C. both A & B
D. none of above

149. Stockton and Johnson" and Higuchi et al. studied
A. the effect of an increase of alcohol concentration on the dissociation constant of sulfathiazole
B. investigated the effect of alcohol on the solubility of phenobarbital
C. both A & B
D. none of above

150. Cliowhan measured and calculated the solubility of the organic carboxylic acid naproxen and its
A. sodium, potassium, calcium, and magnesium salts.
B. cobalt, potassium, calcium, and magnesium salts.
C. lithium, potassium, calcium, and magnesium salts.
D. zinc, potassium, calcium, and magnesium salts.

151. A lower pKa value indicates a
A. stronger base, that is, the lower value indicates the base partially dissociates in water.
B. stronger acid, that is, the lower value indicates the acid more fully dissociates in water.
C. stronger acid, that is, the lower value indicates the acid partially dissociates in water.
D. stronger base, that is, the lower value indicates the acid more fully dissociates in water.

152. Butter et al. demonstrated that phenobarbital,
A. in highly alkaline solution, the second hydrogen ionizes.
B. in weakly alkaline solution, the second hydrogen ionizes.
C. in highly alkaline solution, only one of hydrogen ionizes.
D. in highly alkaline solution, not ionizes the hydrogen

153. Although the barbiturates are soluble in alkalines,
A. they are precipitated as the free acids when a weaker acid is added and the pH of the solution is lowered

B. they are precipitated as the free acids when a stronger acid is added and the pH of the solution is lowered
C. they are precipitated as the free bases when a stronger acid is added and the pH of the solution is lowered
D. they are precipitated as the free acids when a stronger acid is added and the pH of the solution is increased

154. If an excess of liquid or solid is added to a mixture of two immiscible liquids,
A. it will distribute itself be1ween the two phases so that each becomes saturated.
B. it will transfer to one phases so that becomes saturated.
C. it will distribute itself be1ween the two phases so that each becomes unsaturated.
D. it will don't able to distribute itself be1ween the two phases.

155. If the substance is added to the immiscible solvents in an amount insufficient to saturated solu1ions,
A. it will still become distributed between the two layers in a definite pressure.
B. it will still become distributed between the two layers in a definite concentration ratio.
C. it will still become distributed between the two layers in a definite temperature.
D. both A & C.

156. C1 / C2 = K The equilibrium constant, K, is known as
A. the distribution ratio
B. distribution coefficient
C. partition coefficient
D. all of above.

157. Equation C1 / C2 = K , which is known as
A. the solubility law, is strictly applicable only in dilute solutions where activity coefficients can be neglected
B. the distribution law, is strictly applicable only in concentrated solutions where activity coefficients can be neglected
C. the distribution law, is strictly applicable only in dilute solutions where activity coefficients can be neglected
D. the distribution law, is strictly applicable only in dilute solutions where activity coefficients can be higher.

158. By use of this equation $W_n = W(K V_1 / K V_1 + V_2)$,
A. it can be shown that most efficient ex traction results when n is large and V2 is small.
B. it can be shown that most efficient ex traction results when n is small and V2 is small.
C. it can be shown that most efficient ex traction results when n is large and V2 is large.
D. it can be shown that most efficient ex traction results when n is small and V2 is large.

159. when a large number of extraction are carried out with
A. large portions of extracting liquid. The development just described assumes complete immiscibility of the two liquids.
B. small portions of extracting liquid. The development just described assumes complete immiscibility of the two liquids.
C. small portions of extracting liquid. The development just described assumes partial immiscibility of the two liquids.
D. large portions of extracting liquid. The development just described assumes partial immiscibility of the two liquids.

160. Solutions of foods, drugs, and cosmetics are subject to
A. development by the enzymes of microorganisms that act as catalysts in decomposition reactions.
B. oxidation by the enzymes of microorganisms that act as catalysts in decomposition reactions.
C. reduction by the enzymes of microorganisms that act as catalysts in decomposition reactions.
D. deterioration by the enzymes of microorganisms that act as catalysts in decomposition reactions.

161. Sterilization and the addition of chemical preservatives are common methods used in pharmacy to
A. preserve drug solutions against attack by various microorganisms.
B. preserve drug solutions against acids.
C. preserve drug solutions against bases.
D. both B & C.

162. Benzoic acid in the form of its soluble salt, sodium benzoate, is often used as preservation substance
A. because it acts as an oxidant agent.
B. because it acts as a reduce agent..
C. because it produces no injurious effects in humans, when taken internally in small quantities..
D. both A & B.

163. Rahn and Conn showed that the preservative or bacteriostatic action of benzoic acid and similar acids is due
A. almost entirely to the undissociated acid and not to the ionic form.
B. almost entirely to the dissociated acid and to the ionic form.
C. almost entirely to the undissociated acid and to the ionic form.
D. almost entirely to the dissociated acid and not to the ionic form.

164. the yeast Saccharomyces ellipsoids, which grows normally at pH of 2.5 to 7.0 in the presence of strong inorganic acids or salts.
A. ceased to grow in the presence of undissociated benzoic acid when the concentration of the acid reached 25 mg/ 100 mL
B. continue to grow in the presence of undissociated benzoic acid when the concentration of the acid reached 25 mg/ 100 mL
C. ceased to grow in the presence of dissociated benzoic acid when the concentration of the acid reached 25 mg/ 100 mL
D. ceased to grow in the presence of undissociated benzoic acid when the concentration of the acid reached 10 mg/ 100 mL

165. The undissociated molecule, consisting of a large nonpolar portion, is
A. insoluble in the lipoidal membrane of the microorganism and penetrate rapidly.
B. soluble in the lipoidal membrane of the microorganism and penetrate rapidly.
C. soluble in the lipoidal membrane of the microorganism and penetrate with difficulty.
D. soluble in the lipoidal membrane of the microorganism and don't penetrate rapidly.

166. Bacteria in oil-water systems are generally located in
A. the organic phase and at the oil-water interface.
B. the aqueous phase and organic phase
C. the aqueous phase and at the oil-water interface.
D. none of above.

167. the efficacy of a weak acid, such as benzoic acid, as a preservative for these systems is largely a result of
A. the concentration of the undissociated acid in the aqueous phase.
B. the concentration of the dissociated acid in the aqueous phase.
C. the concentration of the undissociated acid in the organic phase.
D. none of above.

168. in equation C = qCo +Cw= q[HA]o +(HA)w + [A-]w
A. q = V /Co
B. q = V /HA
C. q = V /Cw
D. q = V /Vw

169. in equation C = qCo +Cw= q[HA]o +(HA)w + [A-]w
A. C is the original concentration or the acid in the water phase before the aqueous solution is equilibrated with peanut oil.

B. C is the original concentration or the acid in the organic phase after the aqueous solution is equilibrated with peanut oil.
C. C is the original concentration or the acid in the water phase before the aqueous solution is equilibrated with peanut oil.
D. none of above

170. in equation C = qCo +Cw= q[HA]o +(HA)w + [A-]w
A. Co is the mole fraction of the simple undissociated molecules in the oil.
B. Co is the molal concentration of the simple undissociated molecules in the oil.
C. Co is the normal concentration of the simple undissociated molecules in the oil.
D. Co is the molar concentration of the simple undissociated molecules in the oil.

171. in equation C = qCo +Cw= q[HA]o +(HA)w + [A-]w
A. Cw , the normal concentration of benzoic acid in water, is equal to the sum of the two terms [HA]w and[A -]w in this ionizing solvent.
B. Cw , the molar concentration of benzoic acid in water, is equal to the sum of the two terms [HA]w and[A -]w in this ionizing solvent.
C. Cw , the molal concentration of benzoic acid in water, is equal to the sum of the two terms [HA]w and[A -]w in this ionizing solvent.
D. Cw , the mole fraction of benzoic acid in water, is equal to the sum of the two terms [HA]w and[A -]w in this ionizing solvent.
172 The distribution of total benzoic acid among the various species in this system depends upon the A. pressure
B. temperature
C. distribution coefficient, K,the dissociation constant
D. both A & B .

173.Who investigated the effect of alcohol on the solubility of phenobarbital?
A .Edmonson
B .Goyan
C .Higuchi
D . A and B

174.Schwartz determined the solubility of phenytoin as a function of pH and alcohol concentration in various buffer systems and calculated the apparent
A . viscosity
B . dissociation constant
C . solubility
D . all of these

175.Kramer and Flynn examined the solubility of hydrochloride salts of organic bases as a function of

A . pH
B . temperature
C . solvent composition
D . all of these

176.Chowhan measured and calculated the solubility of the organic carboxylic acid naproxen and its sodium, potassium, calcium, and magnesium salts.

A . true
B . false

177.If C1 and C2 are the equilibrium concentrations of the substance in Solvent1 and Solvent2, respectively, the equilibrium expression becomes

A . C1/C2 = K
B . C2/C1 = K
C . C1/C = K
D . C/C2 = K

178.The equilibrium constant, K, is known as

A . distribution ratio
B . distribution coefficient
C . partition coefficient
D . all of these

179. C1/C2 = K This equation is known as

A . distribution ratio
B . distribution law
C . distribution coefficient
D . partition coefficient

180.Wn = w(K v1/ K v1+ v2) By use of this equation, it can be shown that most efficient extraction results when n is large and V2 is small.

A . true
B . false

181.Knowledge of partition is important to the pharmacist because the principle is involved in several areas of current pharmaceutical interest. These include

A . oil–water systems
B . drug action at nonspecific sites
C . the absorption and distribution of drugs throughout the body
D . all of these

182.The concentration of solute remaining in the first solvent is (w1/V1) g/mL and the concentration of the solute in the extracting solvent is (w – w1)/V2 g/mL. The distribution coefficient is thus

A . K = (w1/v1)/(w- w1)v2

B . w1 = w.K v1/ K v1+ v2

C . both of these

D . none of these

183.Distribution law, is strictly applicable only in ____ solutions where activity coefficients can be neglected.

A . dilute

B . concentrated

C . heterogenous

D . homogenous

184.If an excess of liquid or solid is added to a mixture of two immiscible liquids, it will distribute itself between the two phases so that each becomes saturated.

A . true

B . false

185.If the substance is added to the immiscible solvents in an amount insufficient to saturate the solutions, it will still become distributed between the two layers in a indefinite concentration ratio.

A . true

B . false

186.Distribution law is given as

A . C1/C2 = K

B . K = (w1/v1)/(w- w1)v2

C . w1 = w.K v1/ K v1+ v2

D . C2/C1 = K

187. Materials that prevent the initiation and growth of microorganisms in products are known as

A . antioxidants

B . preservatives

C .chelating agents

D . buffers

188.Which of the following are examples of preservatives?

A . Phenethyl Alcohol

B . Potassium Sorbate
C . Propylene Glycol
D . All of these

189.What are the synonyms of Alcohol?
A . Ethanol
B . Ethyl alcohol
C . Grain alcohol
D . All of these

190. H_3BO_3 is molecular formula of
A . Alcohol
B . Benzalkonium chloride
C . Boric acid
D . Sorbic acid

191. Butylated Hydroxyanisole (BHA) is used as
A . antioxidant
B . preservative
C . buffer
D . A andB

192. Butylparaben is used as
A . Antimicrobial preservative for liquid
B . Antimicrobial preservative for semi-solid preparations
C . Both of A & B
D . None of these

193. Phenylethyl Alcohol is used as
A . antioxidant
B . antimicrobial preservative
C . buffer
D . chelating agent